

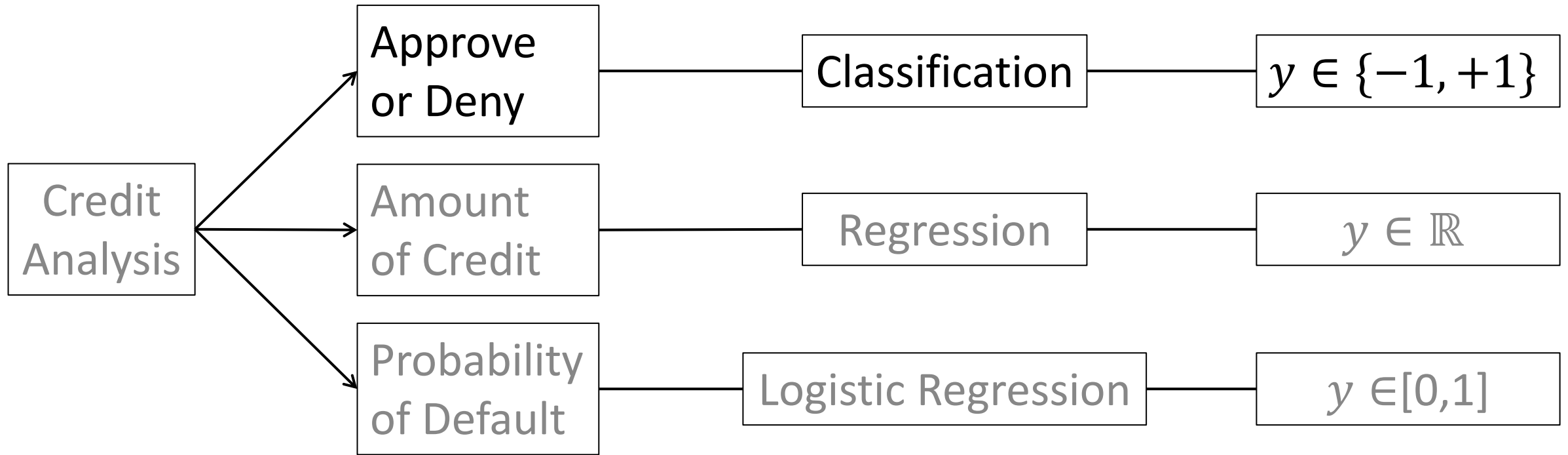


Introduction to Machine Learning (CS 436/536)

Lecture-5: Linear Regression (One Variable)

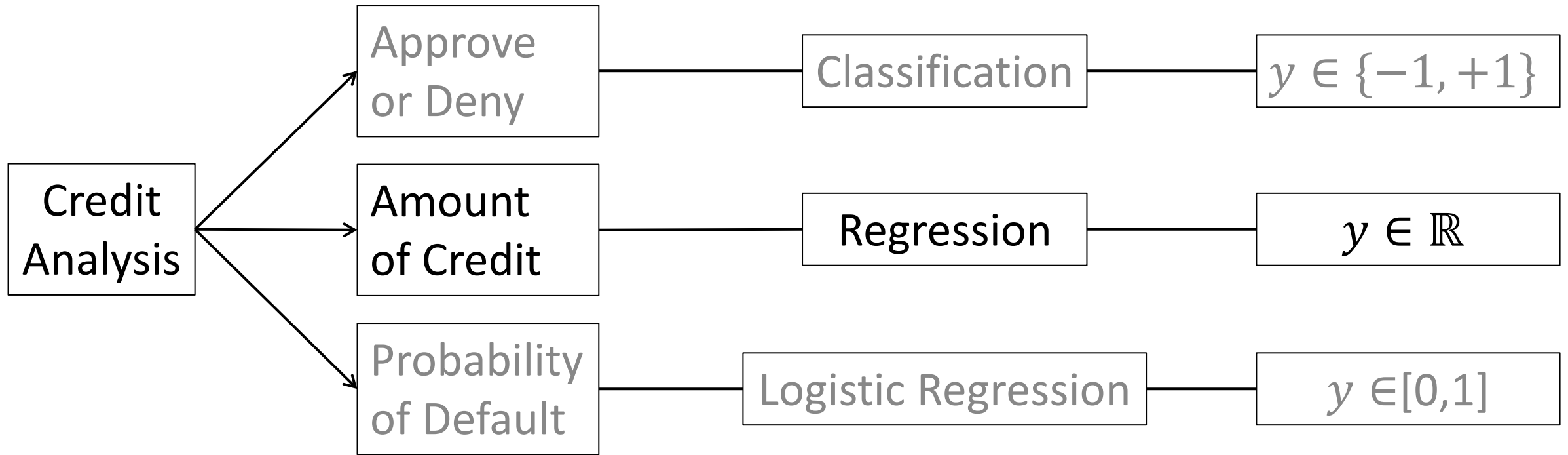
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Linear Models for Three Learning Problems



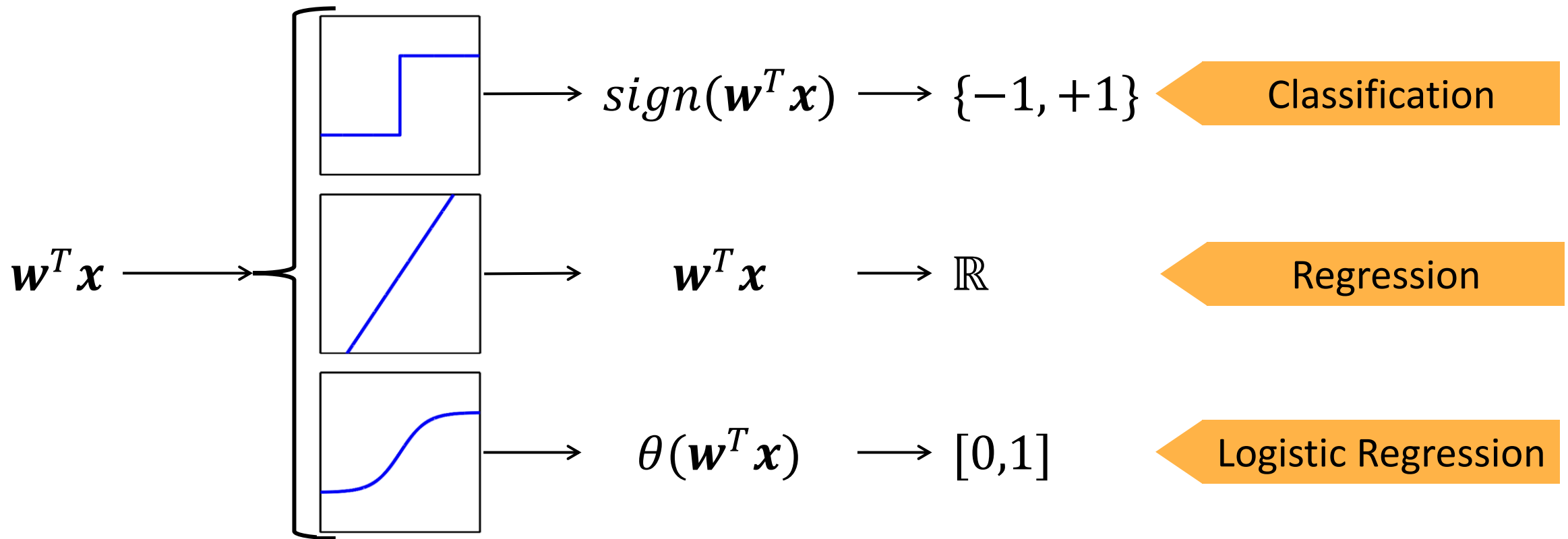
- Fundamental: Building block for more complex models
- First model to try!

Linear Models for Three Learning Problems



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- First model to try!

Linear Models: The Linear Signal



Applications of Regression:

- House prices (?, ?, ?)
- Stock prices (?, ?, ?)
- Exam Scores (?, ?, ?)
- Rain Forecast (?, ?, ?)

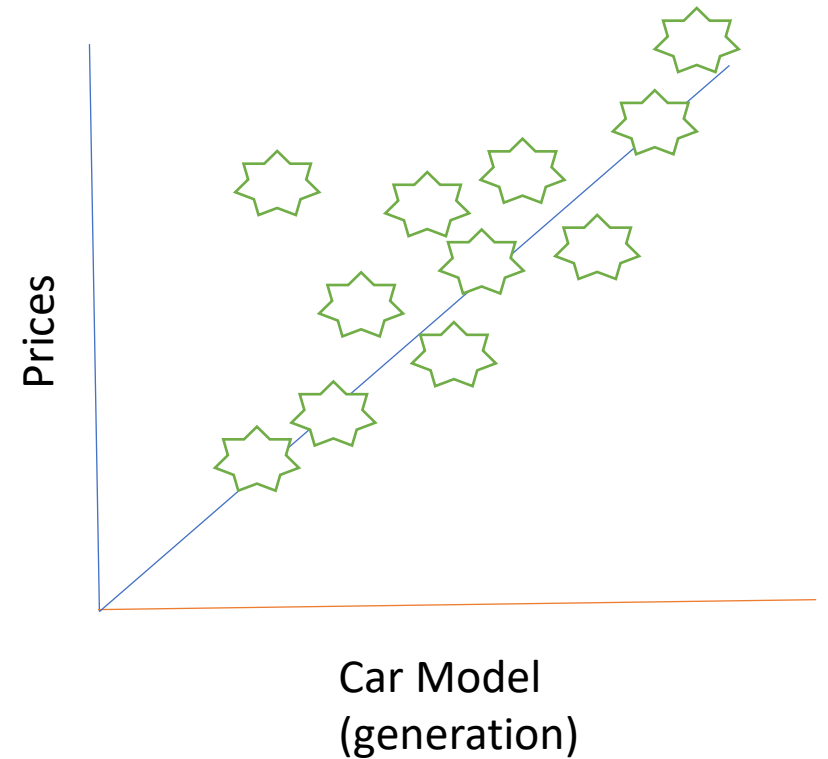
Applications of Regression:

- House prices (Demand, Size, Income)
- Stock prices (Market, Investors, Economy)
- Exam Scores (Course, Student Prep, Teacher)
- Weather Forecast (Season, Humidity, Temperature)

An Example of Regression Problem:

Car generation (x)	Prices (y)
1	4000
2	10000
3	25000

- Training Data (x_i, y_i is the i^{th} training sample)
- Number of training samples m
- Input Features (x)
- Output values (y)

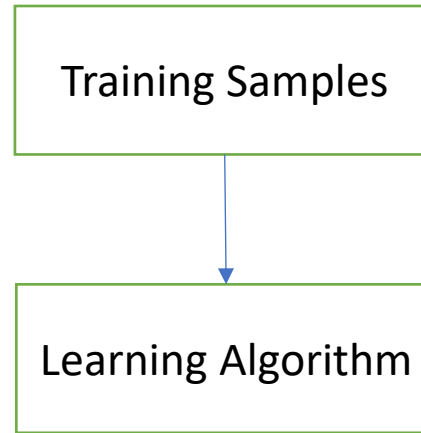


Recap of convolutional supervised learning

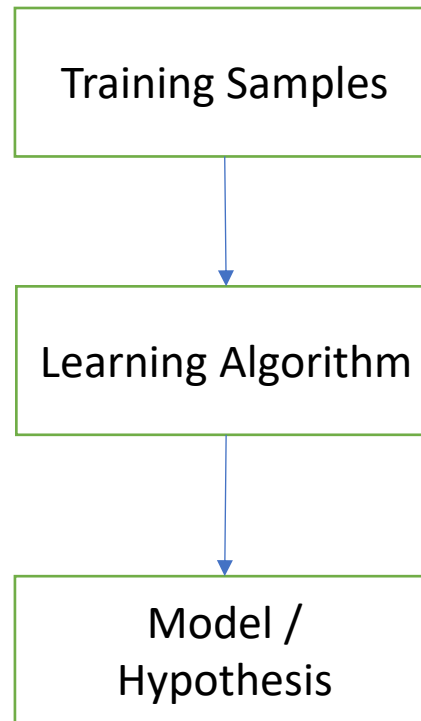


Training Samples

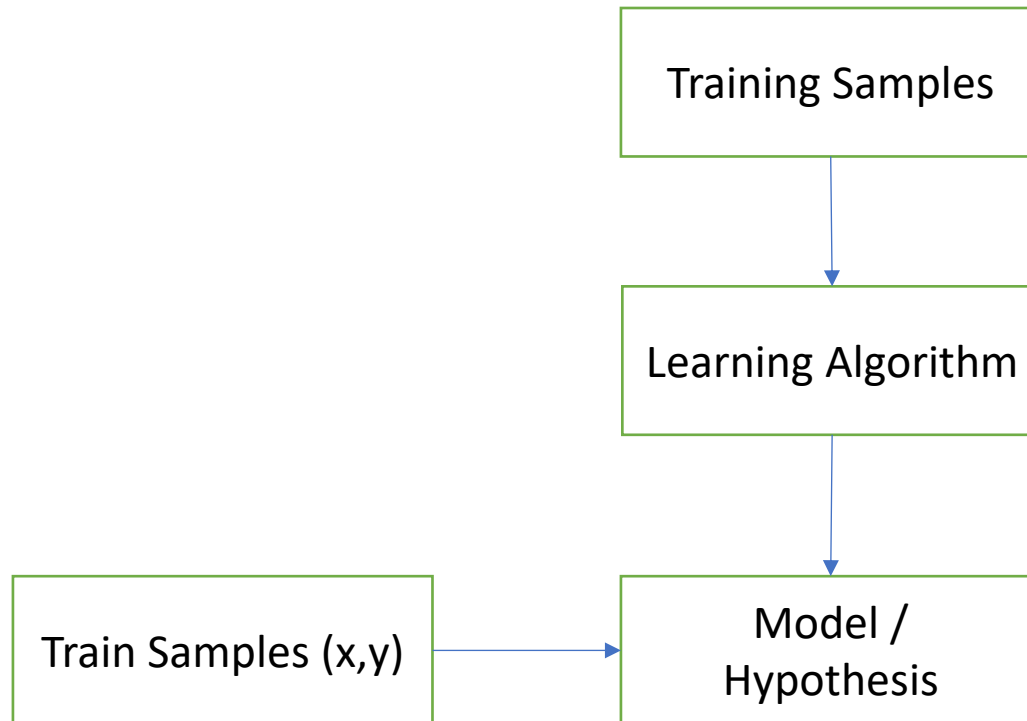
Recap of conventional supervised learning



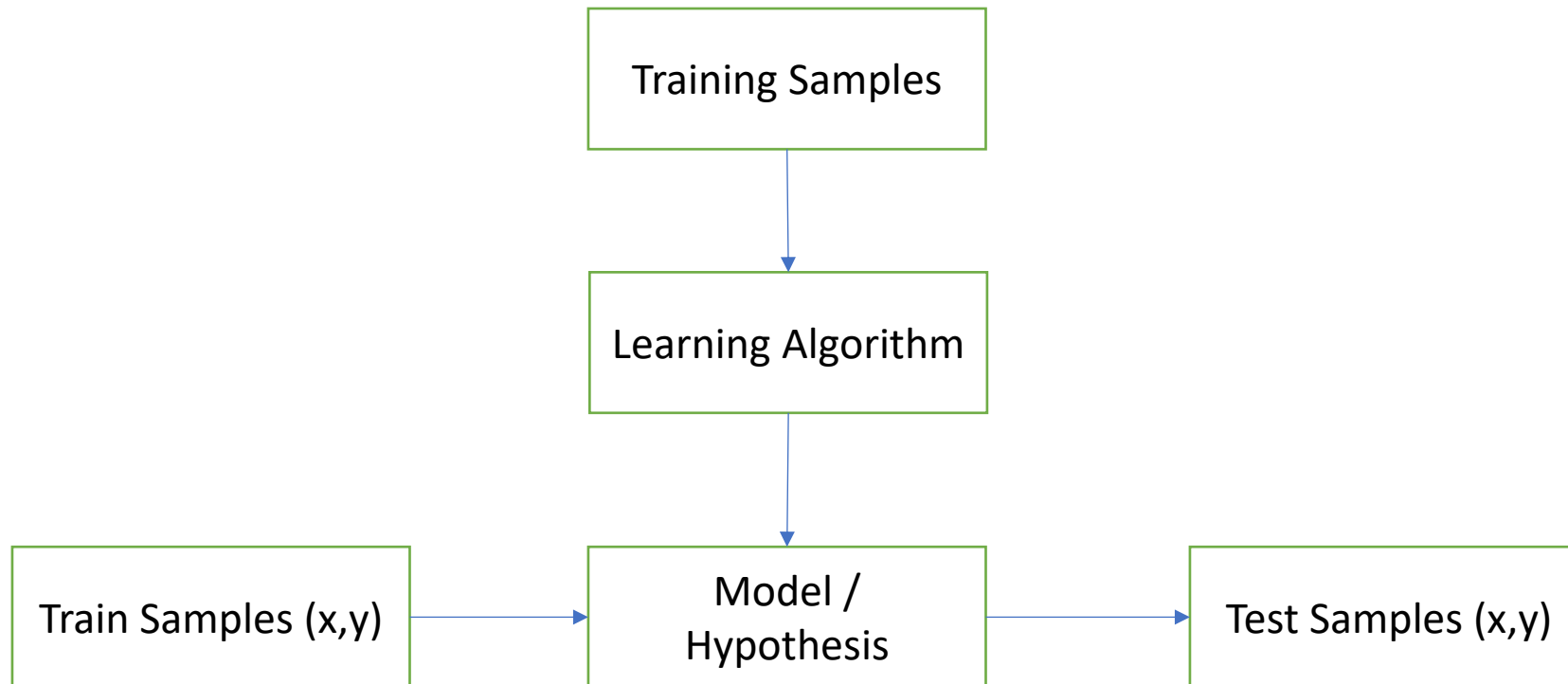
Recap of conventional supervised learning



Recap of Convex Supervised Learning



Recap of convolutional supervised learning



Model Representation

- Model is denoted by $f_{\theta}(x)$ or $f(x)$
- Θ represents the parameter of the regression model.
- A typical Regression expression is $f_{\theta}(x) = \theta_0 + \theta_1 x$
- where θ_0 is the bias parameter
- Formally this assumes that the relationship between x and y is linear.

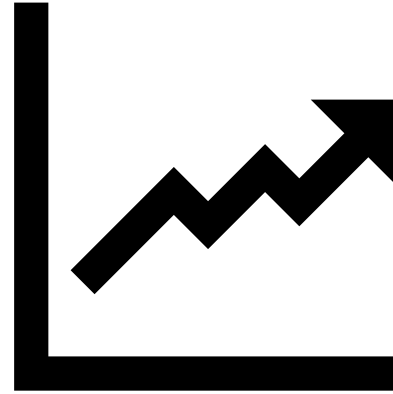
Linear Regression with One Variable

- Model is denoted by $f_{\theta}(x)$ or $f(x)$
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$$f_{\theta}(x) = \theta_0 + \theta_1 x$$

More into the model function

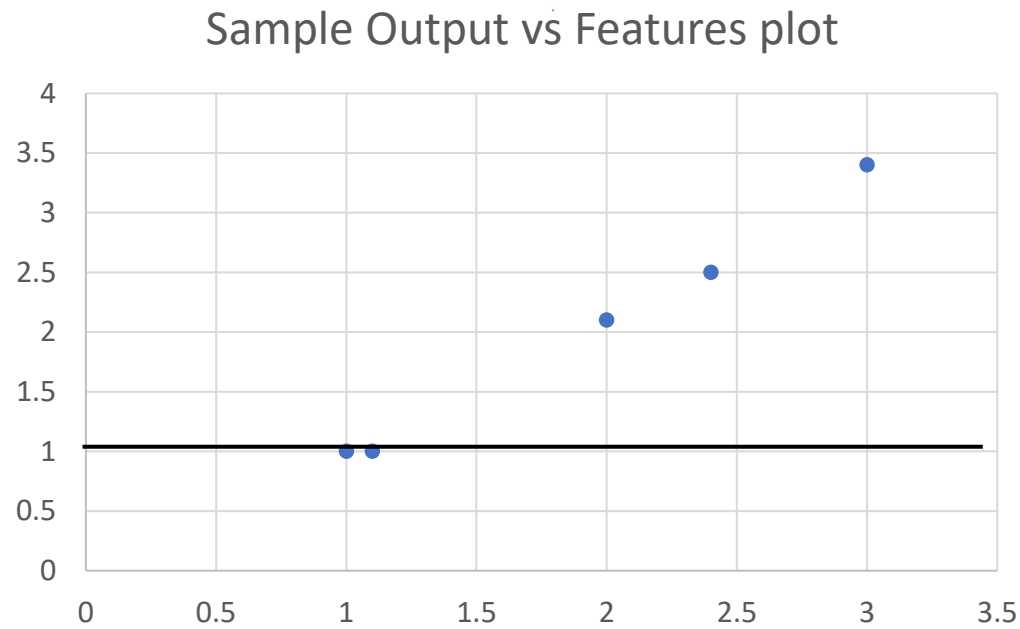
The value of θ_0 and θ_1 decides the type of the function.



More into the model function

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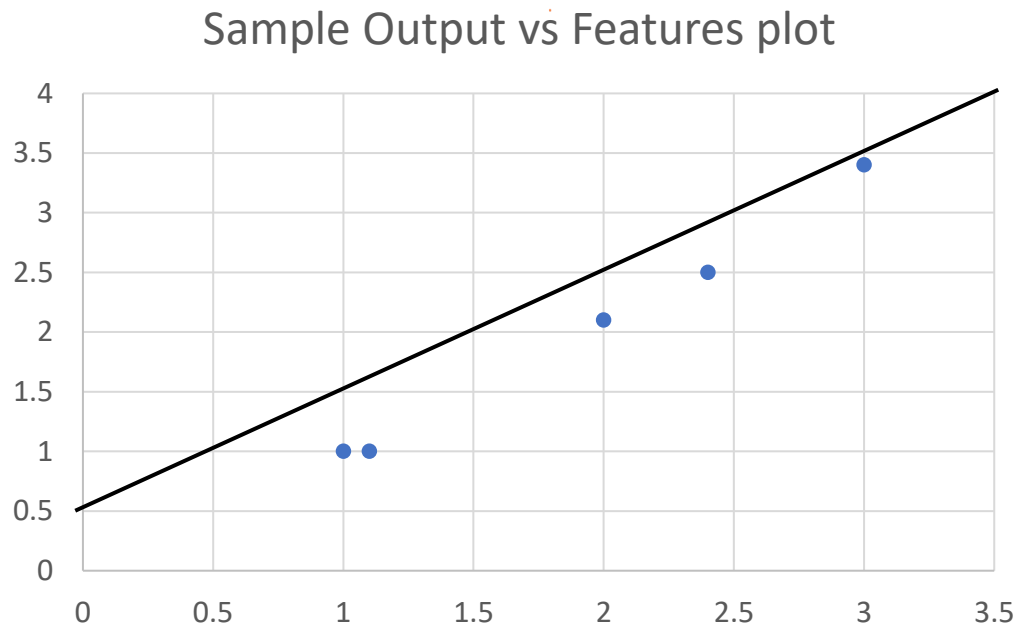
If $\theta_0 = 1$ and $\theta_1 = 0$:



More into the model function

The value of θ_0 and θ_1 decides the type of the function.

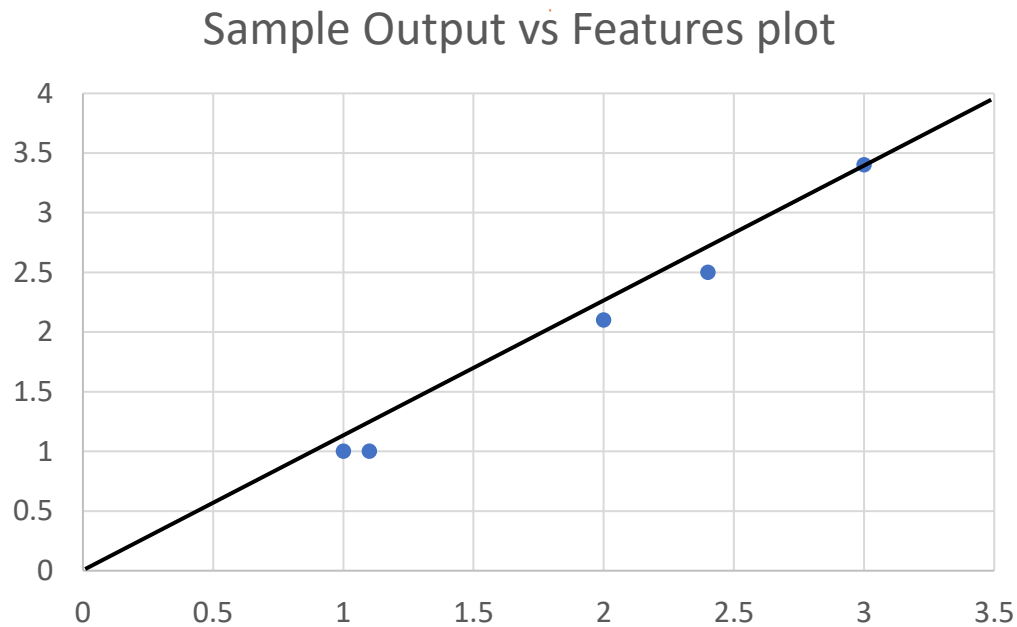
If $\theta_0 = 0.5$ and $\theta_1 = 1$:



More into the model function

The value of θ_0 and θ_1 decides the type of the function.

If $\theta_0 = 0$ and $\theta_1 = 1$:



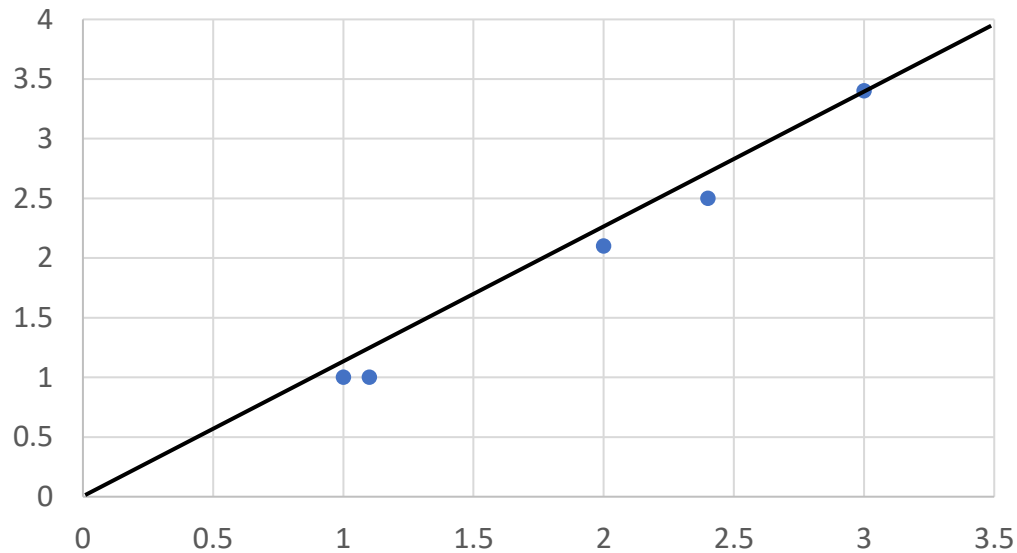
More into the model function

The value of θ_0 and θ_1 decides the type of the function.

If $\theta_0 = 0$ and $\theta_1 = 1$:

Overall Objective of the learning scheme is to select θ_0 and θ_1 Such that the output $f(x)$ closely matches with y .

Sample Output vs Features plot

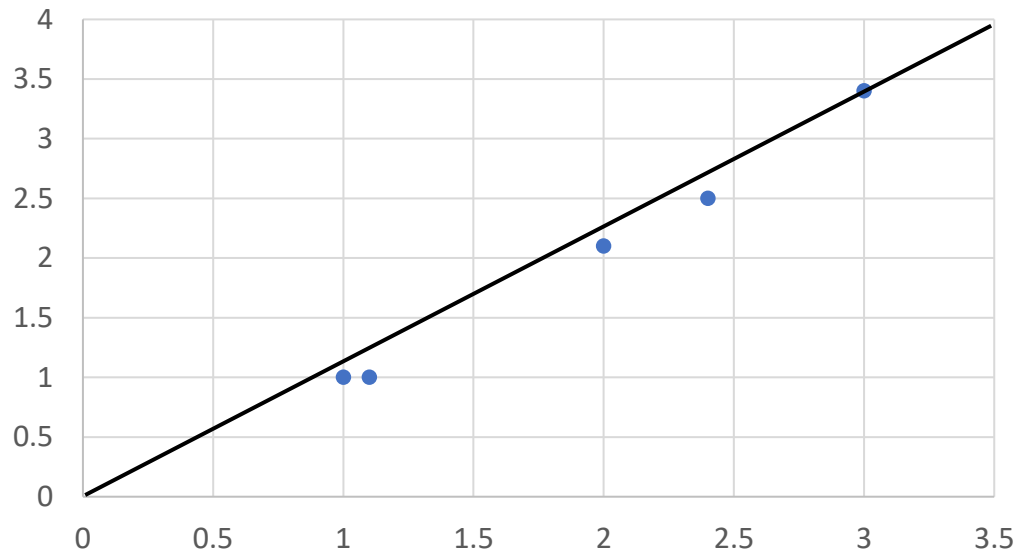


More into the model function

The value of θ_0 and θ_1 decides the type of the function.

If $\theta_0 = 0$ and $\theta_1 = 1$:

Sample Output vs Features plot



Overall Objective of the learning scheme is to select θ_0 and θ_1 Such that the output $f(x)$ closely matches with y .

How to quantify this error?

Objective of the Learning Algorithm

- Mean squared Error:

$$\frac{1}{2m} \sum_i (f(x_i) - y_i)^2$$

- Objective Function of the training algorithm:

$$\underset{\theta_0, \theta_1}{\text{minimize}} J(\theta_0, \theta_1) = \underset{\theta_0, \theta_1}{\text{minimize}} \left(\frac{1}{2m} \sum_i (f(x_i) - y_i)^2 \right)$$

To Summarize

- We have a model function : $f_{\theta}(x) = \theta_0 + \theta_1 x$
- Parameters: θ_0 , θ_1
- Cost Function: $J(\theta_0, \theta_1) = \left(\frac{1}{2m}\right) \sum_i (f(x_i) - y_i)^2$
- Goal : minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

General Algorithm

1. Initialize the model parameters θ_0 , θ_1
2. Compute Output $f(x)$ and Objective function J
3. Update $\theta_j = \theta_j - \alpha \frac{\delta J}{\delta \theta_j}$, $j = 0, 1$; α is the learning rate
4. Compute Output $f(x)$ and Objective function J ; until J is minimized.

Mathematical Expressions

- $\frac{\delta J}{\delta \theta_0} = ??$

- $\frac{\delta J}{\delta \theta_1} = ??$

$$J(\theta_0, \theta_1) = \left(\frac{1}{2m}\right) \sum_i (f(x_i) - (y_i))^2$$

$$f_{\theta}(x) = \theta_0 + \theta_1 x$$

Putting the value of $f(x)$ in the loss function then differentiating to minimize:

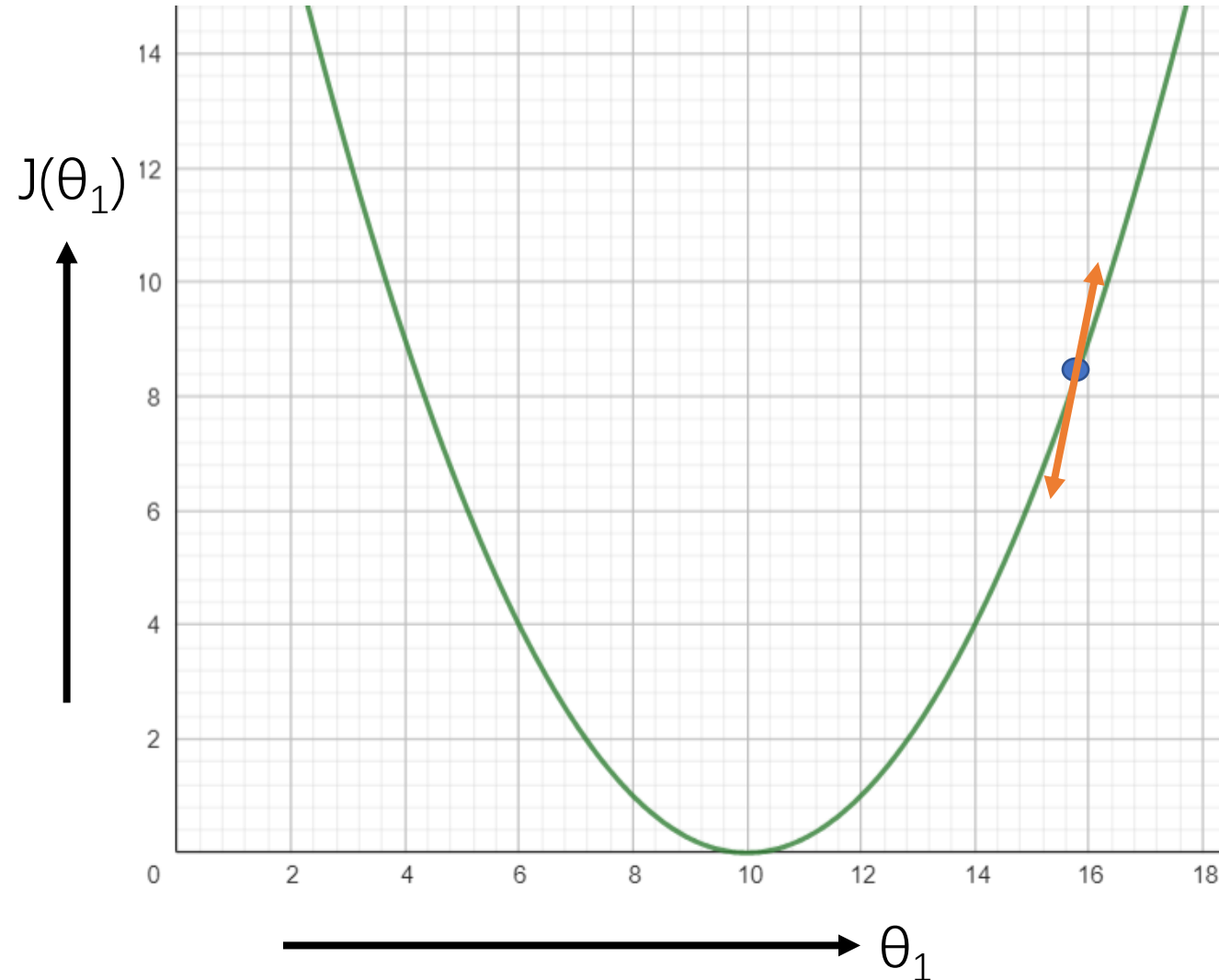
By chain rule:

$$\frac{\delta J}{\delta \theta_0} = \frac{\delta J}{\delta f(x)} \cdot \frac{\delta f(x)}{\delta \theta_0} \Rightarrow \frac{1}{m} \sum_i (f(x_i) - (y_i)) \cdot 1 \Rightarrow \left(\frac{1}{m} \sum_i (f(x_i) - (y_i)) \right)$$

Similarly:

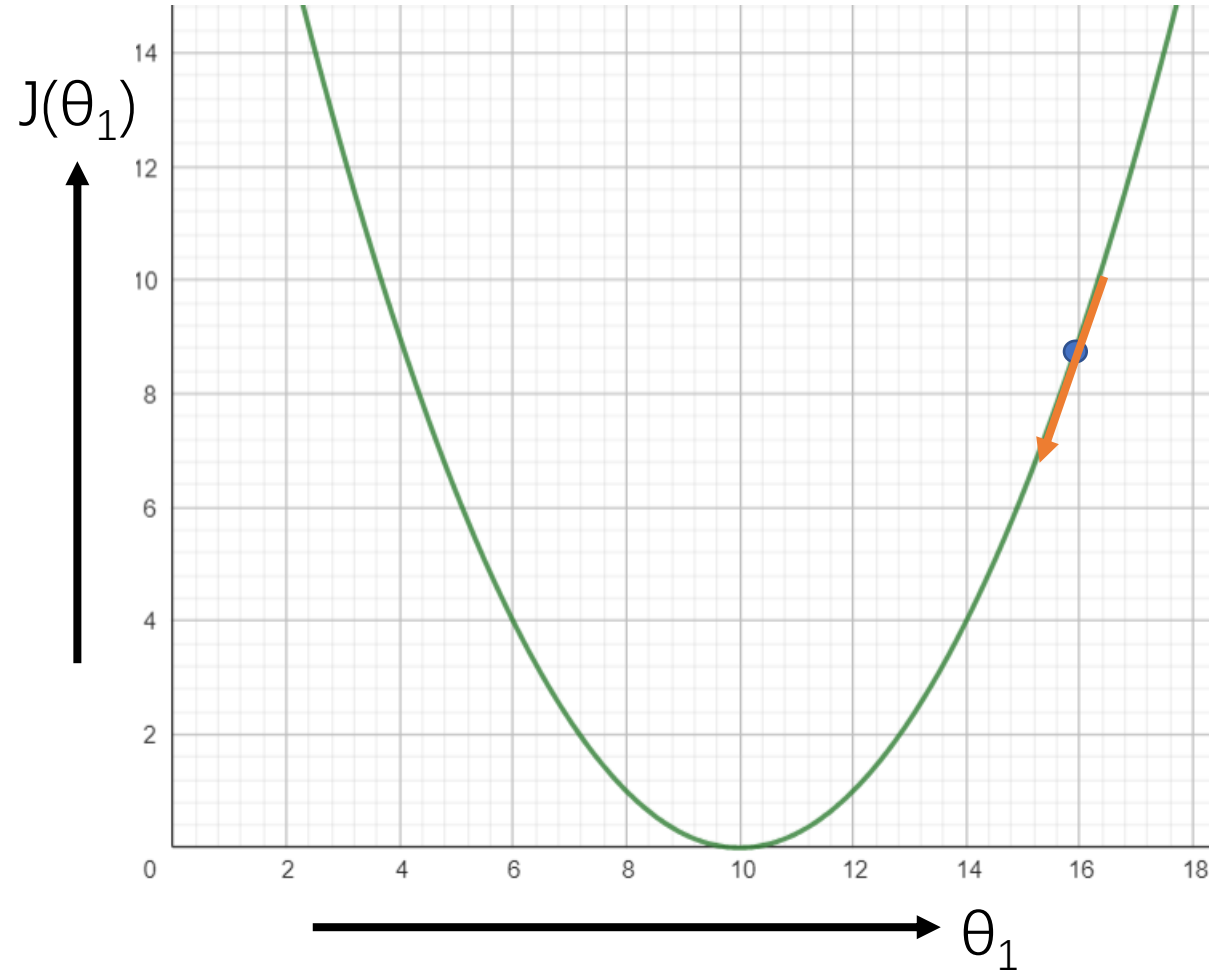
$$\frac{\delta J}{\delta \theta_1} = \frac{\delta J}{\delta f(x)} \cdot \frac{\delta f(x)}{\delta \theta_1} \Rightarrow \frac{1}{m} \sum_i (f(x_i) - (y_i)) \cdot x \Rightarrow \left(\frac{1}{m} \sum_i (f(x_i) - (y_i)) \cdot x \right)$$

Intuition of How the Learning Algorithm works:



$$\theta_1 = \theta_1 - \alpha \frac{\delta J}{\delta \theta_1}$$

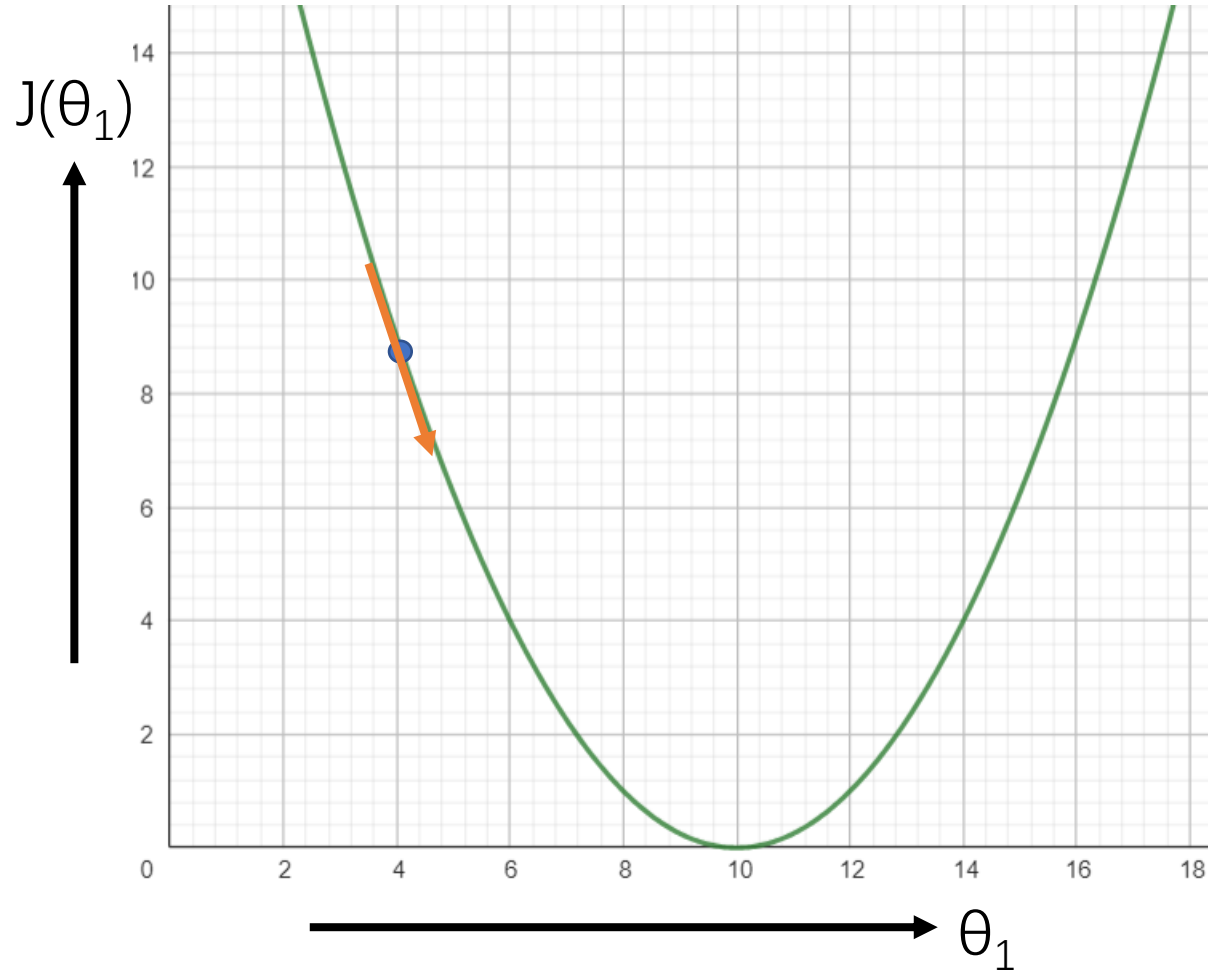
Intuition of How the Learning Algorithm works:



$$\theta_1 = \theta_1 - \alpha \frac{\delta J}{\delta \theta_1}$$

$$\theta_1 = \theta_1 - \alpha(+)$$

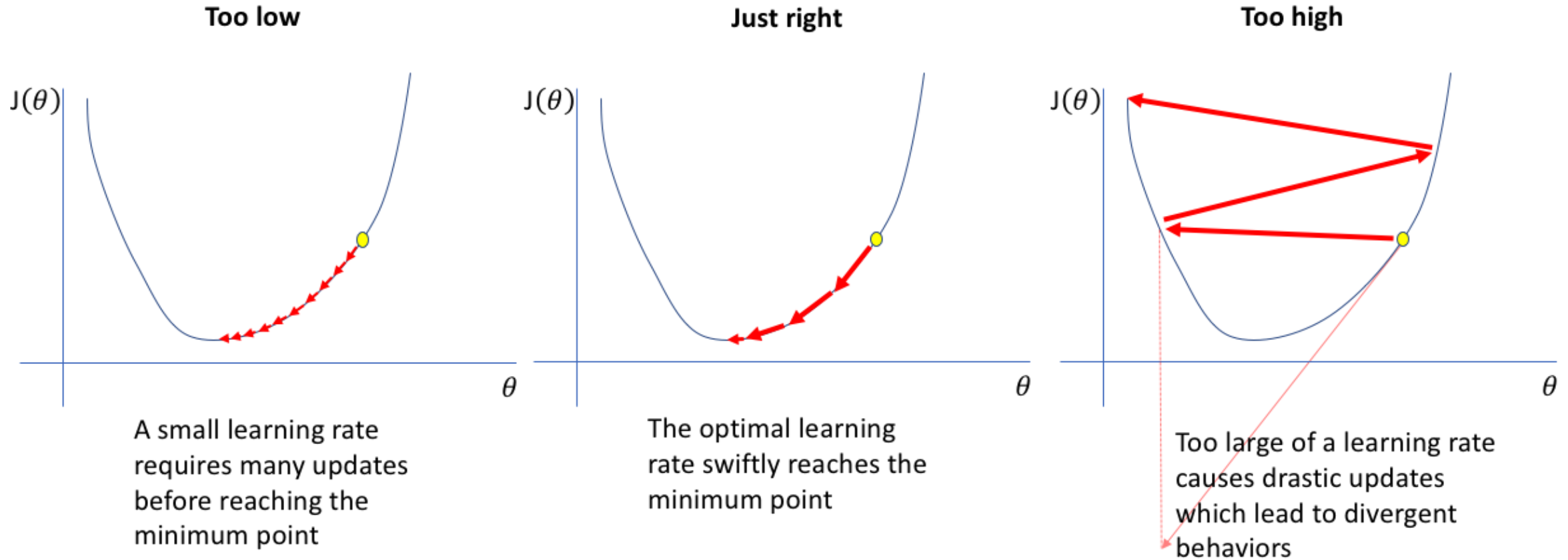
Intuition of How the Learning Algorithm works:



$$\theta_1 = \theta_1 - \alpha \frac{\delta J}{\delta \theta_1}$$

$$\theta_1 = \theta_1 - \alpha(-)$$

Impact of the Learning Rate:



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4. Compute Output $f(x)$ and Objective function J ; until J is minimized.

Working Example:

Area	Price(\$)
160	400
210	523
300	719

Assume Learning rate = 0.001, $\theta_0 = 100$, $\theta_1 = 2$

- Output (for i=1) $\Rightarrow 2*160 + 100 \Rightarrow 420$; Error = $(420 - 400)^2 \Rightarrow 400$
- Output (for i=2) $\Rightarrow 2*210 + 100 \Rightarrow 520$; Error = $(520 - 523)^2 \Rightarrow 9$
- Output (for i=3) $\Rightarrow 2*300 + 100 \Rightarrow 700$; Error = $(700 - 719)^2 \Rightarrow 361$
- Compute Gradient $\frac{\delta J}{\delta \theta_0} = 1/3((420 - 400) + (520 - 523) + (700 - 719)) = -0.66$
- Compute Gradient $\frac{\delta J}{\delta \theta_1} = 1/3((420 - 400)*160 + (520 - 523)*210 + (700 - 719)*300) = -1043.33$
- New $\theta_0 = 100 + 0.001 * (0.66) = 100.00066$
- New $\theta_1 = 2 + 0.001 * (1043.33) = 3.04$

Working Example:

Car Generation (X)	Price (1000) (Y)
1	1
2	2
3	4

- Assume Learning rate = 0.1, $\theta_0 = 0.5$, $\theta_1 = 0.5$
- Output (for i=1) = 1; Error = 0
- Output (for i=2) = 1.5; Error = 0.25
- Output (for i=3) = 2; Error = 4
- Compute Gradient $\frac{\delta J}{\delta \theta_0} = 1/3(0 - 0.5 - 2) = -0.833$
- Compute Gradient $\frac{\delta J}{\delta \theta_1} = 1/3(0 - 0.5*2 - 2*3) = -2.333$
- New $\theta_0 = 0.5 + 0.1 * 0.833 = 0.5833$ and $\theta_1 = 0.5 + 0.1 * 2.333 = 0.7333$

Working Example:

Car Age (X)	Price (1000) (Y)
1	1
2	2
3	4

What is the impact of the update?

- Output (for $i=1$) = 1.3166; Error = 0.1
- Output (for $i=2$) = 2.0499; Error = 0.002
- Output (for $i=3$) = 2.7832; Error = 1.48
- Price of the car (Next generation) = 3.52

References and Further Reading

- Andrew Ng's Coursera lecture [Linear Regression— Andrew Ng – YouTube](#)
- *FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS. Algorithms, Worked Examples, and Case Studies* John by D. Kelleher Brian Mac Namee Aoife D'Arcy (Permission taken MIT Press)
- *Dive into Deep Learning*, Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola (July 25, 2021)
- Regression Paper Review: [5 2019 07 06!06 49 29 PM.pdf \(uomustansiriyah.edu.iq\)](#)